Inventory Tracker and Extraction Tool – Luis Weekes

Architecture and approach:
My overall approach was for the LLM to extract raw values and then for deterministic rules to be used to fill in ditto marks and to highlight risky rows. The final output needs to be assessed and be approved by a human—they can change values and exclude entire rows if necessary. For this project, the risk tolerance is very low by default—you will see many medium risk values and quite a few high ones.

Users upload handwritten inventory sheets as PDFs or images through a Streamlit frontend, where the documents are normalised and passed to an OpenAI Vision LLM for extraction into structured JSON using Pydantic schemas. The deterministic layer then, resolves contextual dependencies which can later be changed given a stake holder's values—dependencies include things such as ditto marks, downward arrows, inherited info. A validation layer then identifies missing values, inconsistencies etc and finally assigns a risk level given certain rules.

The architecture deliberately separates probabilistic AI usage from deterministic business rules and human approval.

Edge Cases and Failure Modes:
Single lines have been seen to pass through the system "|" and are not always recorded as inherited values. Many rows are seen as medium risk, but rules can be changed. Highlights are automatically flagged as high risk although not always true—it is debatable whether this is a feature or a bug.

A Realistic Target for Production and How to Evaluate
A realistic target for production would be to have runs and values persisted in a real backend and a real deployed front end. For a system like this to go out for production, we need to verify with many more different inventory slip samples and to create an automated testing rubric with ground truth values, once we reach a designated accuracy level, we could do a first deployment.

HITL for low-confidence cases:
I would extend on my current design where rows are flagged with risk levels and also assign an overall file risk level. I would extend the explanations for why something was labelled with what risk level. I would design a system where if there is a row that is truly high risk, the original inventory slip writer would be automatically contacted.

Assumptions
- Blank cells and ditto values continue the value above
- An invalid date inherits the valid date above
- Risk follows fixed thresholds after inheritance
- A person needs to approve the whole sheet